The Agentic Paradigm Shift

The Agentic Paradigm Shift

Agents don’t finish your sentences - they finish your goals by planning steps, using tools, and verifying results.

Why it Matters

- Plan → Act → Verify
- Tools to prove changes work
- Human-in-the-loop

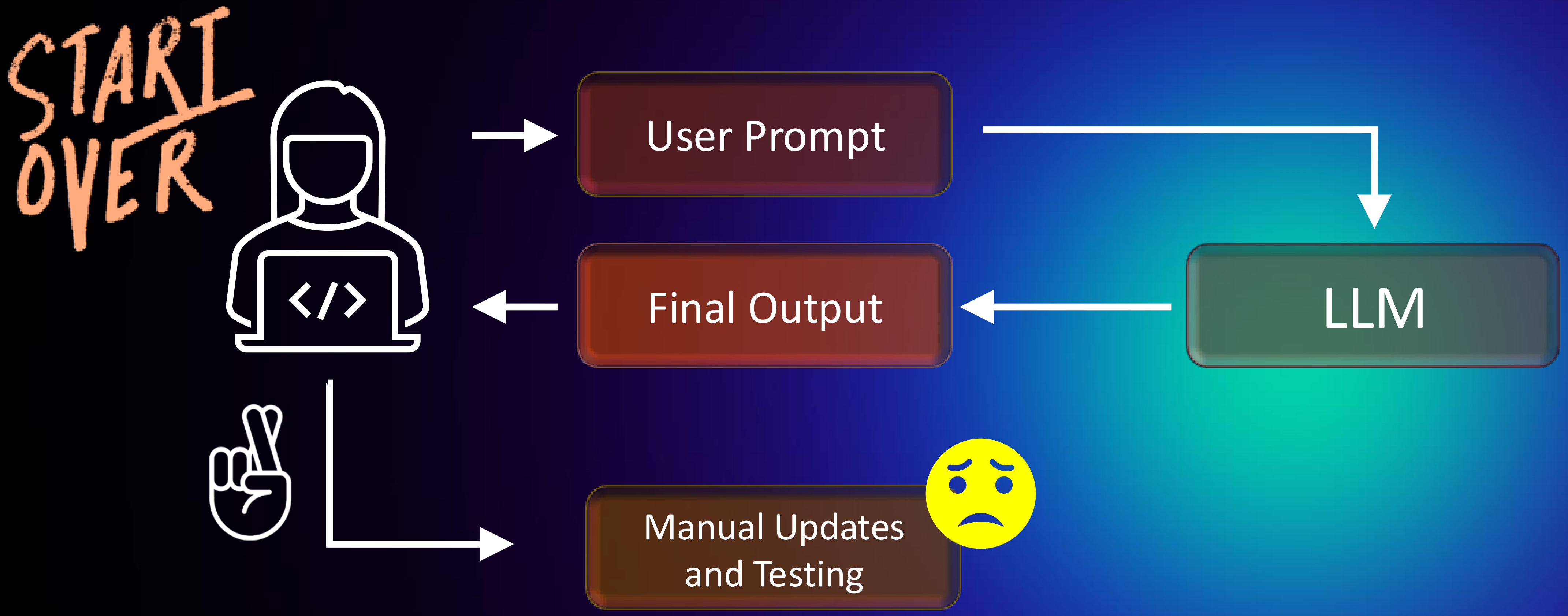
The Old Model

- Token-by-token suggestions
- One-off snippets, no execution
- You verify manually
- Easy drift & rework

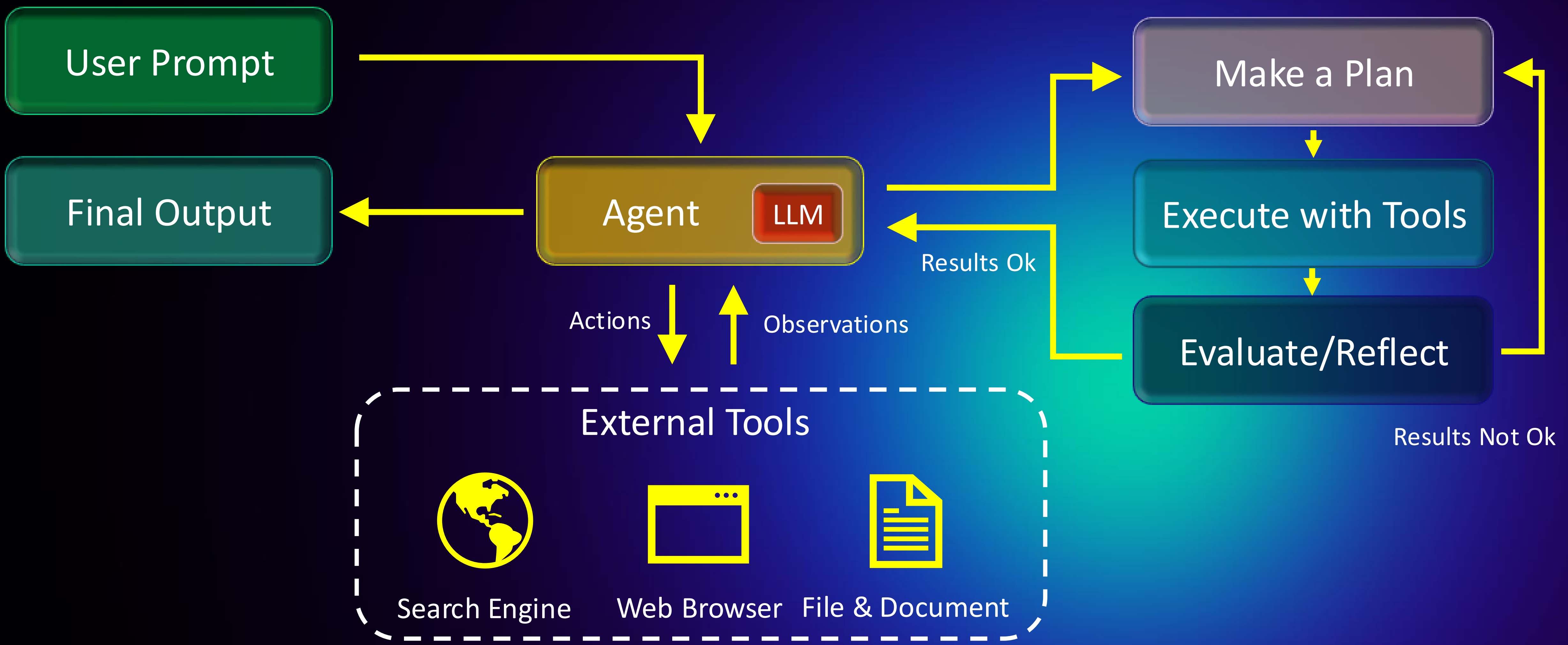
Agentic Model (Coding Agent / Agent Mode)

- Goal → plan → multi-step run
- Calls tools (test, build, refactor)
- Diffs + checkpoints for review
- Repeatable tasks, less toil

Pre-Agentic Workflow: Prompt-In, Output-Out



The Agentic AI Agent Workflow



What is an Agent?

AN LLM-powered controller that plans, calls tools, observes results, and replans until done.

The Controller

- Plans steps, select tools, executes
- Uses context & memory to iterate and improve
- Evaluates results - replans until done

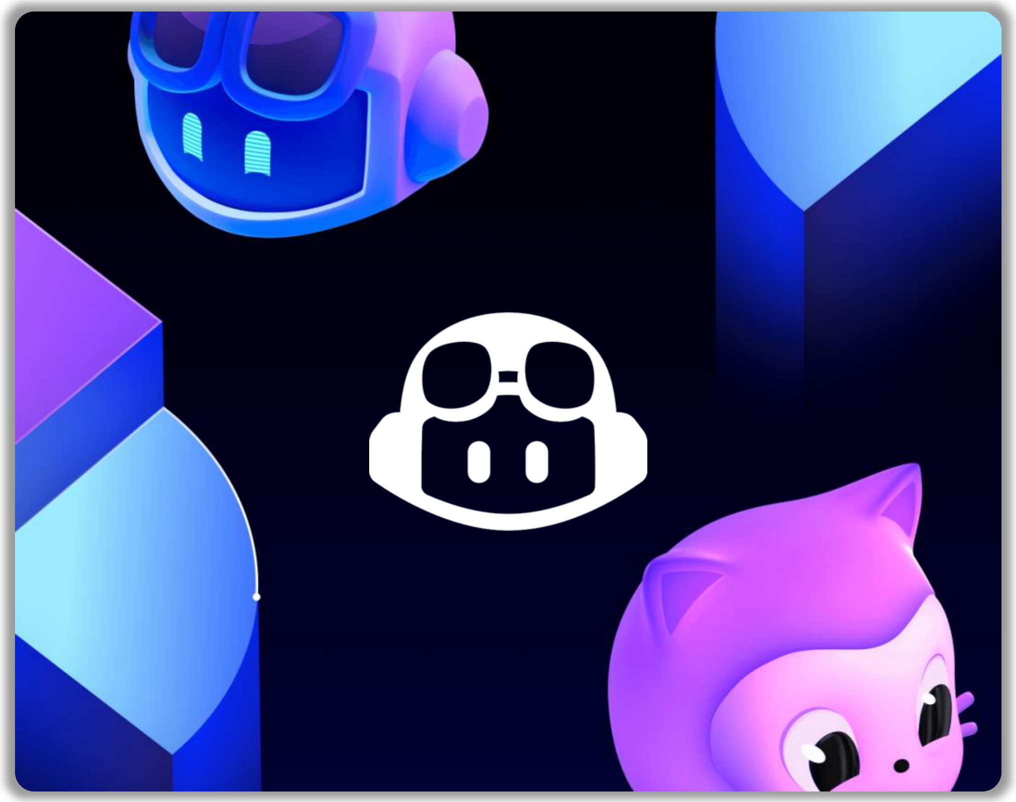
Definition of Done

- Objective checks pass (tests, linters, build)
- Reproducible trace (plan, commands, diffs)
- Acceptance criteria met and summary generated

Human-in-the-Loop

- Approve/deny steps and modify the plan
- Provide domain context and constraints
- Override/stop actions; request explanations

What is the GitHub Copilot Coding Agent?



What is the GitHub Copilot Coding Agent?

The GitHub Copilot Coding Agent is an autonomous AI developer that works independently in the background to complete larger, multi-step tasks assigned through GitHub issues, chat prompts, or comments.

GitHub Copilot Coding Agent

Lives In The Cloud

Given an issue, will develop, run tests/linters, and open a PR in the background

Repo-Aware

Uses repository context to make changes and presents clean diffs, summaries, and PRs

Create and Execute End-to-End Tests

Explore code, make changes, executes automated tests in an ephemeral development environment

How It Works (At a Glance)

Define the Task
Open an issue or prompt with context, constraints, and a clear Definition of Done

Agent Runs in the Cloud
Plans steps, edits code across files, runs tests/linters/ build, retries on failures.

You Review and Ship
Inspect diffs, logs, and the PR summary; request changes or approve and merge.

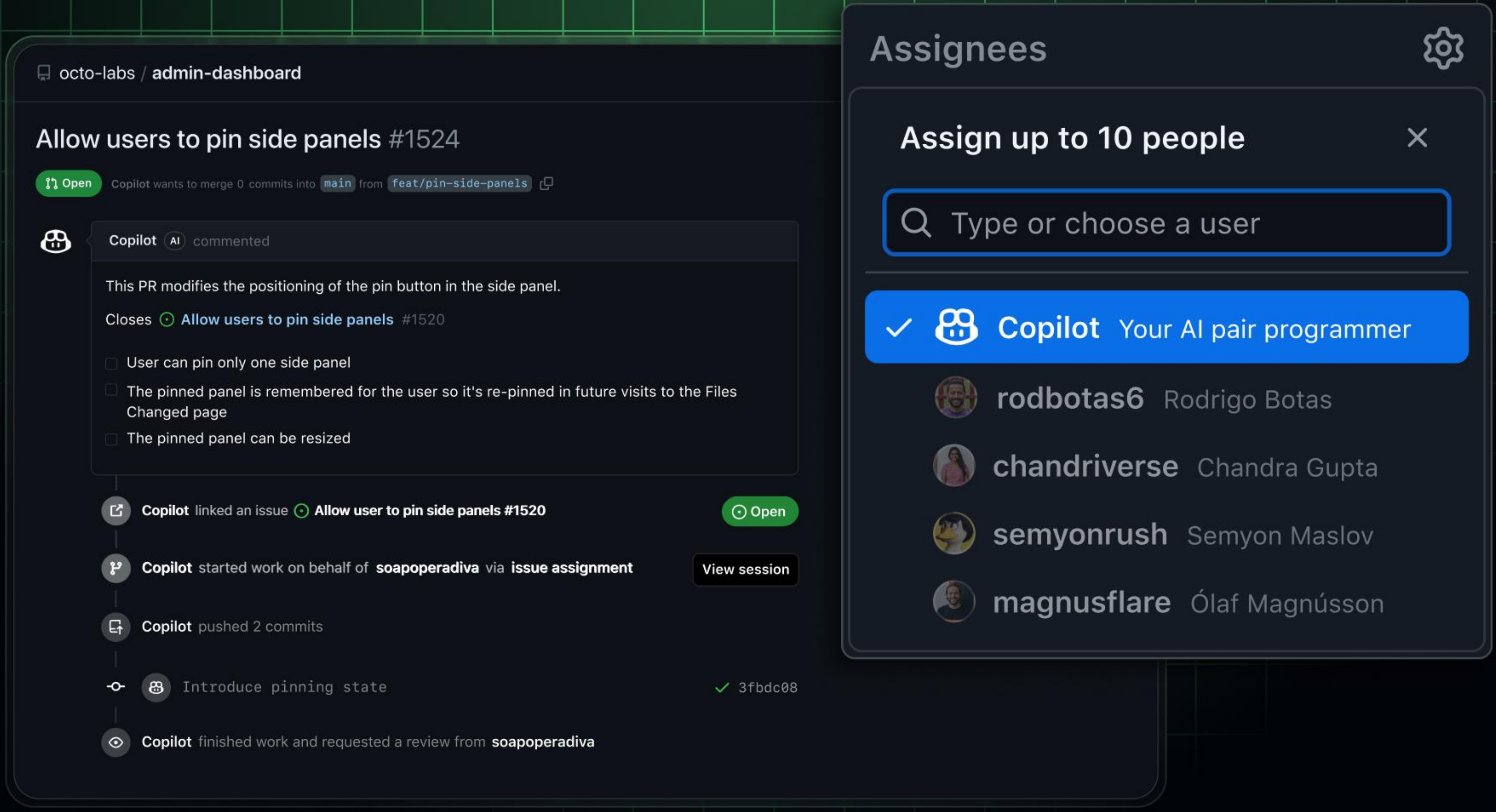


Image Source: github.blog

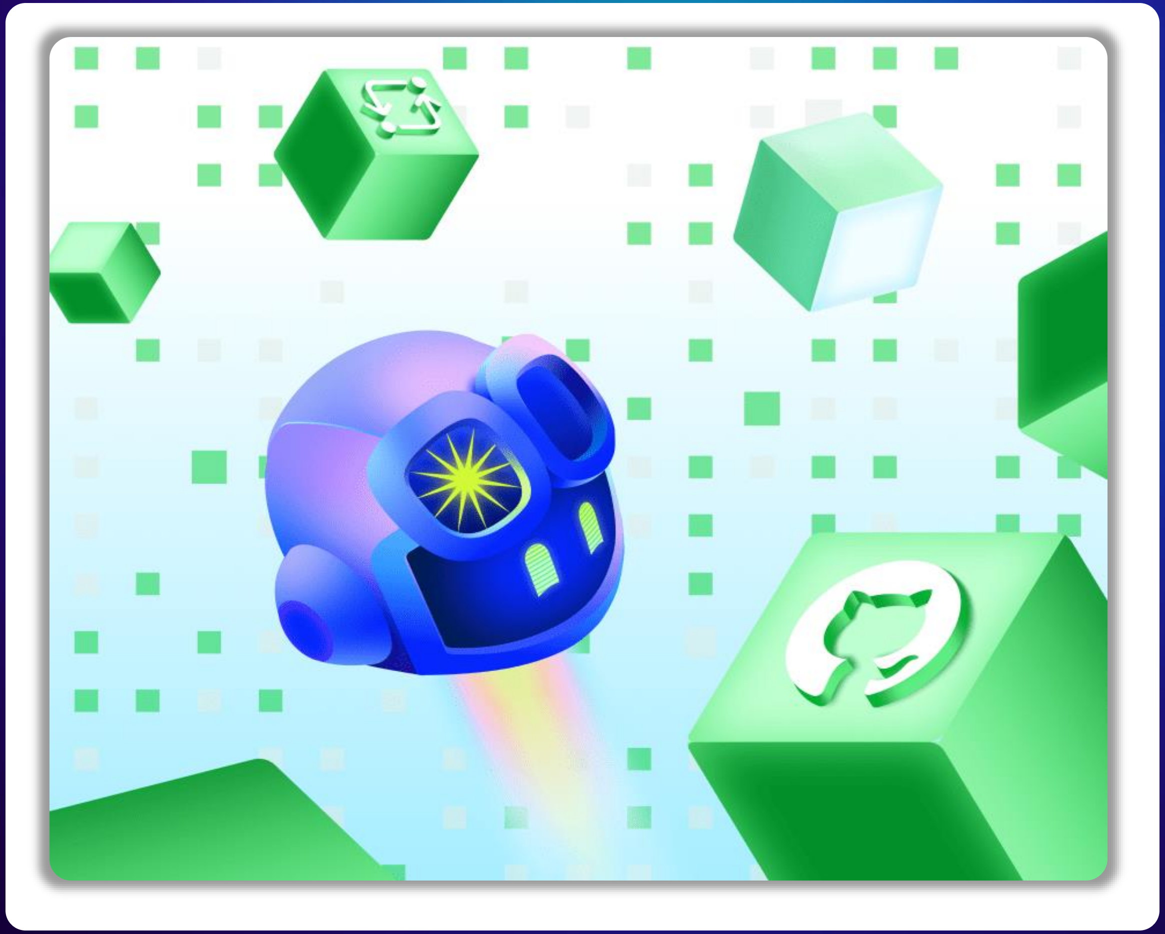
Prerequisites, Access Requirements, & Limitations

GitHub
Subscription Tier

The Agent is a Premium Feature

The Coding Agent is not included with the standard individual Copilot plan.

- GitHub Copilot Pro
- GitHub Copilot Pro+
- GitHub Copilot Business
- GitHub Copilot Enterprise



Cost Considerations

The agent uses Copilot premium requests and consumes GitHub Actions minutes, which may impact your budgeting.

Each plan includes a fixed number of premium requests per user per month. You can purchase additional premium requests for $0.04 each.

Pro Tip: Set a budget to receive alerts when you reach 75%, 90%, or 100% of your budget.

Coding Agent: Boundaries

Copilot coding agent has certain limitations in its software development workflow and compatibility with other features.

Same Repository

Copilot can only make changes in the same repository where it is creating its pull request.

Limted Context

Copilot can only access context in the same repository as the assigned issue.

One Pull Request

Copilot can only open one pull request at a time.

No Existing PRs

Copilot cannot work on an existing pull request that it dic create.

Coding Agent Limitations

Important Limitations to Be Aware of When Working with the Coding Agent

Unsigned Commits

Commits aren’t signed so signed-commit branches will require history rewrite

Content Exclusions Ignored

Agent doesn’t honor Copilot content-exclusion rules.

Cannot Use Self-Hosted Runners

Runs only on GitHub-hosted runners so you can't use your self-hosted runners

GitHub-Only Repos

Works only on repositories hosted on GitHub and not a locally hosted GitHub Enterprise

No Managed-User Personal Repos

GitHub-hosted runners are not available to repositories owned by managed user accounts

No LLM Model Choice

You cannot switch between or pick a model to use – GitHub might change at any time.

Coding Agent vs. Agent Mode

“Give a dev a code completion and they’ll merge once.

Teach a dev to wield an AI agent and they’ll empty the backlog before the coffee cools.“

- GitHub

An Asynchronous Teammate that Lives and Develops In The Cloud

Lives within your GitHub repos and can take on issues assigned to it. It will edit code in its own branch, run tests for validation, and open a pull request so you can review before merging.

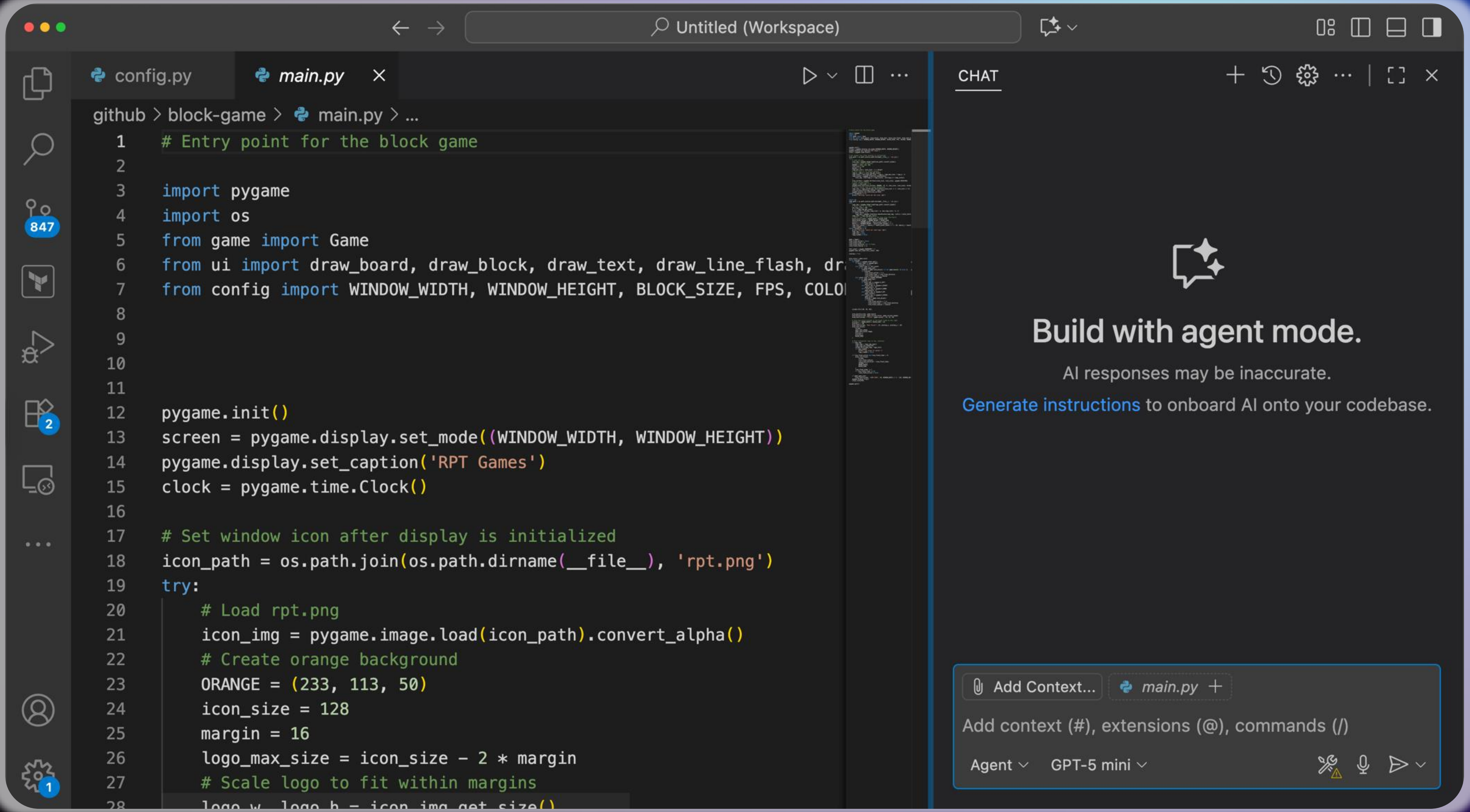
VS.

A Real-Time Collaborator that Sits Directly in Your Editor

Works inside of VS Code that can provide code suggestions, run tests, and answer questions given a natural-language goal. All changes are made locally.

Pro Tip: You can, and should, use both agents!

Agent Mode



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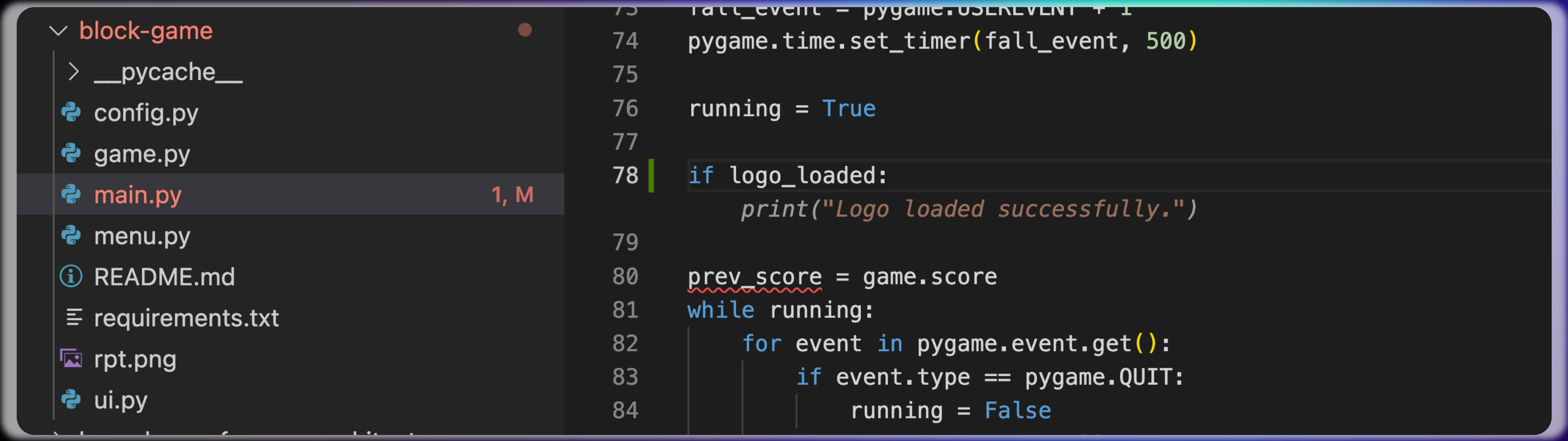
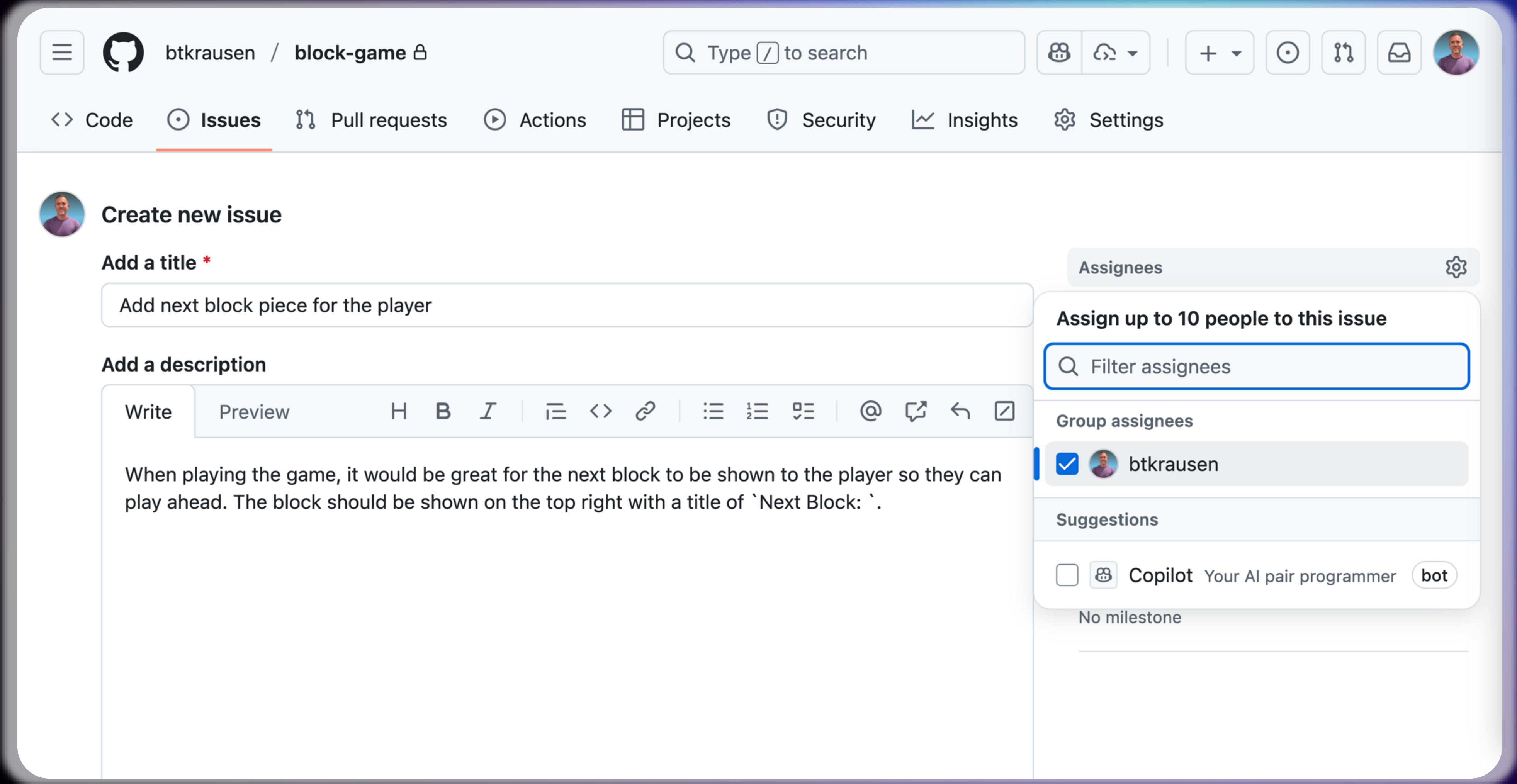
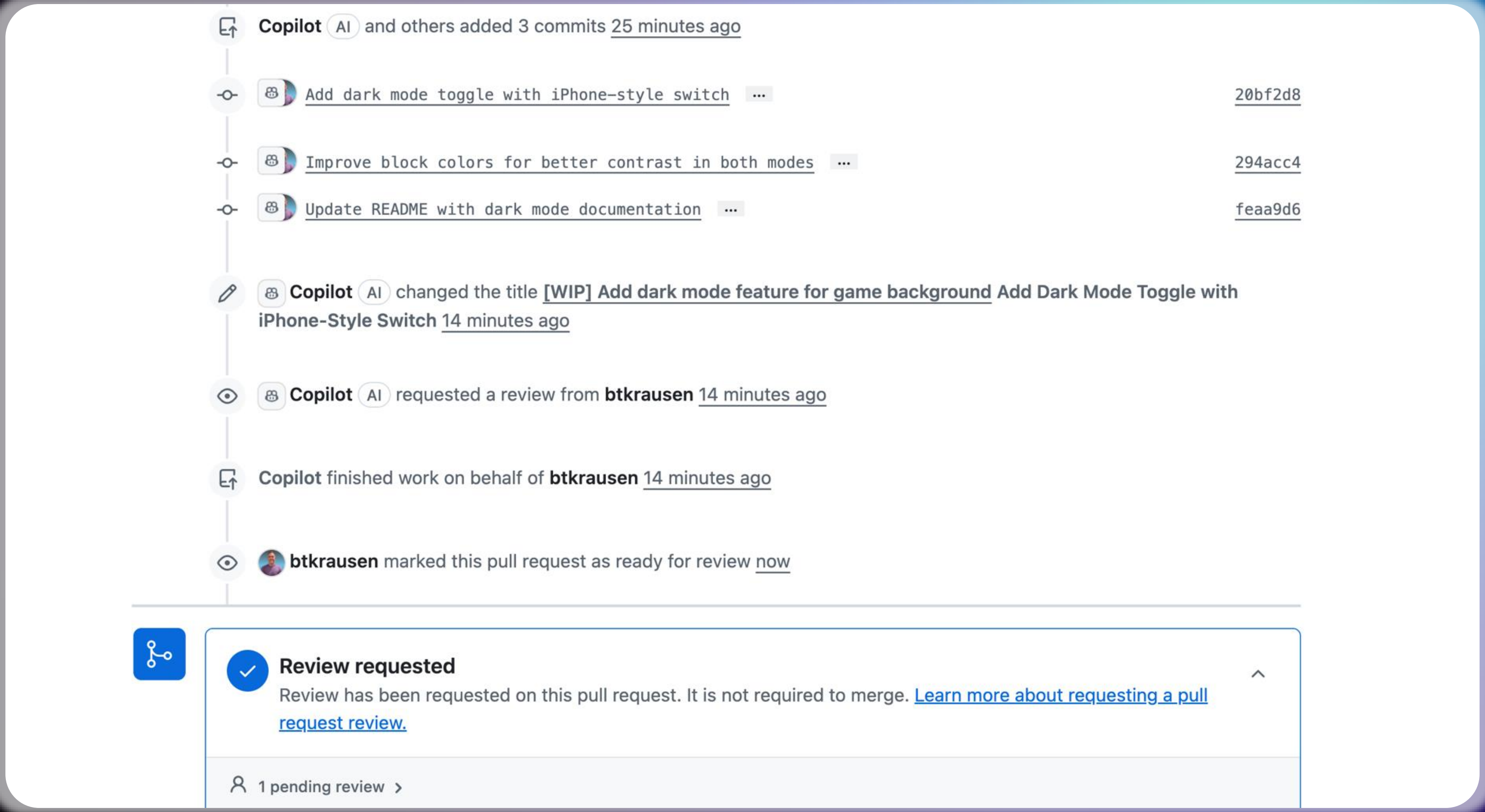
Agent Mode


block-game
__pycache__
config.py
game.py
main.py 1, M
menu.py
README.md
requirements.txt
rpt.png
ui.py
pygame.time.set_timer(fall_event, 500)
running = True
if logo_loaded:
print("Logo loaded successfully.")
prev_score = game.score
while running:
for event in pygame.event.get():
if event.type == pygame.QUIT:
running = False

Coding Agent



Coding Agent



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Coding Agent vs. Agent Mode

Summary

GitHub Copilot Agent Mode

Lives in your editor for autocompletions, chat, file editing, and more.

GitHub Copilot Coding Agent

Lives in the cloud for working assigned issues, testing, and creating pull requests.

DEMO:

Coding Agent

THANK YOU

INSTRUCTOR

Bryan Krausen